



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

Main University Examinations, Academic Year: 2025/2026

**SECOND YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
MECHATRONIC ENGINEERING**

EMT 3201: PRODUCTION TECHNOLOGY

8.30AM-10.30AM

DATE: 3RD February, 2026

TIME: 2 Hours

INSTRUCTIONS

1. This paper contains **FIVE (5)** questions.
2. Question **ONE** is compulsory.
3. Answer Question **ONE** and any other **TWO** questions.
4. Question **ONE** carries **30 marks**, the rest carry **20 marks** each.

QUESTION ONE (COMPULSORY) (30 Marks)

- A. Using suitable sketches, differentiate between the following methods of fabricating plastics.
- i. Blow molding and compression molding [4 Marks]
 - ii. Injection molding and extrusion [4 Marks]
- B. Glass is produced by heating the raw materials to an elevated temperature above which melting occurs. This enables the use of forming processes to produce the desired shape. With aid of suitable diagram describe two glass forming processes. [6 Marks]
- C. Using suitable illustrations, discuss the difference between Isostatic pressing and tape casting process in the fabrication of crystalline ceramics. [6 Marks]
- D. Casting is a solidification process where the molten material is poured into a mold and then allowed to freeze into the desired final shape. Cooling curves are useful tools for studying the solidification process. With aid of a well labelled diagram, discuss the major features and terminologies used in a eutectic-composition alloy cooling curve. [10 Marks]

QUESTION TWO (20 Marks)

A. With aid of suitable sketches, show the metal forming processes and steps involved in the manufacturing of a two piece aluminium beverage can. [10 Marks]

B. Estimate the force required in punching a 30mm diameter hole through a 5mm thick annealed titanium sheet at room temperature. Assume the UTS for Titanium to be 1000 MPa. [3

Marks]

C. With aid of a well labeled sketches, discuss the following operations:

- i. Dinking
- ii. Lancing
- iii. Piercing
- iv. Blanking

[4 Marks]

D. Describe the following methods of making sand molds:

- i. Jolting
- ii. Match-plate molding

[3 Marks]

QUESTION THREE (20 Marks)

A. Shell molding, also known as shell-mold casting is an expendable mold casting process that uses resin covered sand to form the mold allowing the production of both ferrous and non-ferrous metal parts for diverse applications. Use illustrations to describe the basic steps in shell molding and state its advantages [10 Marks]

B. How do you calculate the minimum bend radius for a specific sheet metal material and thickness? What factors or properties of the material are crucial in this calculation? [2 Marks]

C. What are the consequences of using a bend radius smaller than the calculated minimum for a given sheet metal? How does this affect the material and the overall quality of the bent part? [2 Marks]

D. With aid of diagrams differentiate between wiper bending and die bending stating an advantage of each process [4 Marks]

E. Briefly describe the following allowances made in casting:

- i. Shrinkage allowance
- ii. Draft allowance

[2 Marks]

QUESTION FOUR (20 Marks)

- A. Use the appropriate sketches to differentiate between centrifuging and true centrifugal casting. [4 Marks]
- B. In casting experiments performed using a certain alloy and type of sand mold, it took 155 seconds for a cube-shaped casting to solidify. The cube measured 50 mm in length. [2 Marks]
- Determine the value C_m of the mold constant using Chvorinov's Rule.
 - For the same alloy and mold type, find the total solidification time TTS for a cylindrical casting in which the radius $r = 15$ mm and height $h = 50$ mm. [4 Marks]
- C. With the aid of a diagram, describe the following terms as used in bending:
- Neutral axis
 - Bend allowance [4 Marks]
 - Minimum bend radius
- D. With the aid of diagrams describe three types of patterns used in sand casting. [6 Marks]

QUESTION FIVE (20 Marks)

- A. Discuss the following terms as used in metal forming processes:
- Recrystallization temperature [2 Marks]
 - Strain hardening
- B. The recrystallization temperature range during hot working depends on different factors, discuss. *Grain size, nature, strain hardening, strain hardening rate* [6 Marks]
- C. What are the key metal characteristics that significantly influence the success of forming operations in sheet metal fabrication? [6 Marks]
- D. Using examples differentiate between Shearing, bending and drawing as methods of metal forming. [6 Marks]